

CPEN 416 / EECE 5710

## **Lecture 14: QPE and order finding**

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Tuesday 28 October 2025

# Announcements

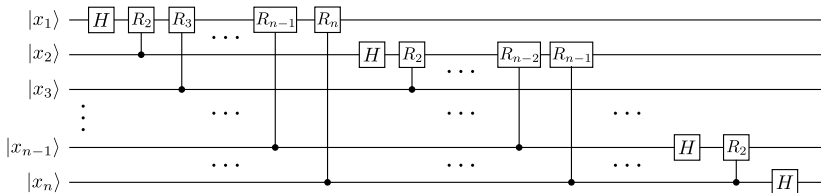
- Quiz 6 today
- Tutorial Assignment 3 due Friday 23:59
- Midterm checkpoint report due Friday **5:00pm**
  - Calendar bookings for checkpoints meetings available Thursday after class
- Practice problem set 3 available (about QPE)

*★ Tomorrow's office hour online*

# Last time

We implemented the quantum Fourier transform using a *polynomial* number of gates and depth.

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$

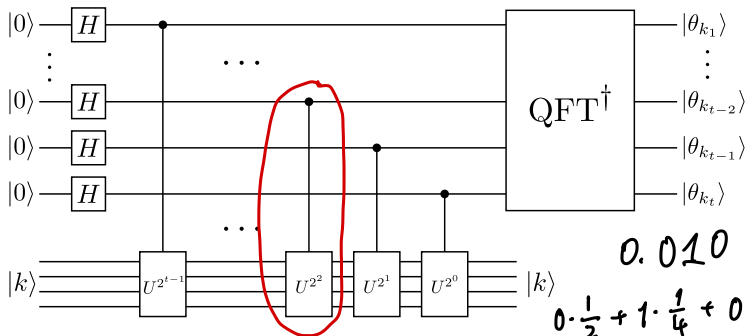


# Last time

We applied the QFT to quantum phase estimation (QPE) to estimate eigenvalues of unitary matrices:

$$U|k\rangle = e^{2\pi i \theta_k} |k\rangle$$

$$U|01\rangle = e^{i\pi/2} |01\rangle$$



$$0.01011$$

$$0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{8} + \dots$$

$$= 0.25 + 0.0625 + \dots$$

$$= 0.3125$$

$$U|01\rangle = e^{i\pi/2}|01\rangle$$

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|01\rangle \longrightarrow \frac{1}{\sqrt{2}}|0\rangle|01\rangle + \frac{1}{\sqrt{2}}|1\rangle(e^{i\pi/2}|01\rangle)$$

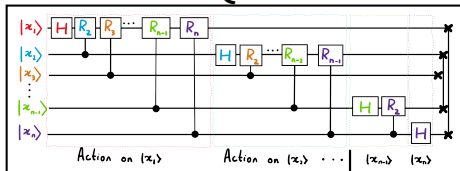
$$= \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/2}|1\rangle)|01\rangle$$

$$e^{2\pi i \cdot 0.????}$$

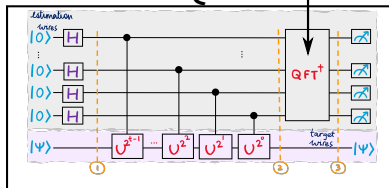
$$e^{-2\pi i - \frac{1}{4}} = e^{2\pi i \cdot 0.01\frac{1}{\infty}}$$

# Reminder: where are we going?

## 1. QFT



## 2. QPE



## 3. Shor

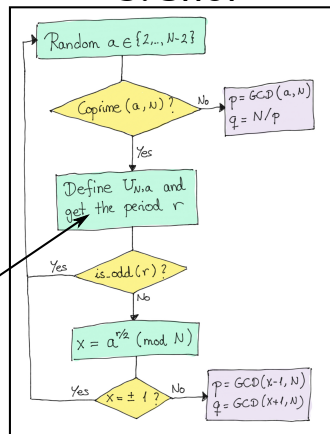


Image credit: Xanadu Quantum Codebook nodes F.3, P.3, S.4

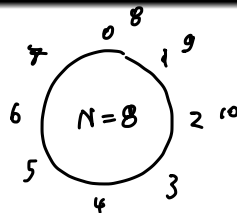
# Learning outcomes

- Perform arithmetic modulo  $N$  with integers and with quantum operations
- Define the order finding problem
- Apply QPE to solve the order finding problem

# Modular arithmetic

Consider the finite set of integers  $\{0, 1, \dots, N-1\}$ .

Basic arithmetic modulo  $N$ :



Addition:  $a + b$        $4 + 6 = 10 \equiv 2 \pmod{8}$

Multiplication:  $4 \cdot 6 \pmod{8} = 24 \equiv 0 \pmod{8}$

# Modular inverses

*multiplicative*

The inverse of  $a$  modulo  $N$  ( $a^{-1}$ ) is an integer  $b$  such that

Regular  $a \rightarrow$  inverse  $\frac{1}{a} \Rightarrow a \cdot b \equiv 1 \pmod{N}$

$$b = a^{-1} \neq \frac{1}{a} \in \text{not ap int.}!$$

A modular inverse only exists when  $a$  and  $N$  are coprime.

Example:  $N = 5$   $2^{-1} \equiv 3 \pmod{5}$

| x | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 | 3 | 4 |
| 2 | 0 | 2 | 4 | 1 | 3 |
| 3 | 0 | 3 | 1 | 4 | 2 |
| 4 | 0 | 4 | 3 | 2 | 1 |

Example:  $N = 4$

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 | 3 |
| 2 | 0 | 2 | 0 | 2 |
| 3 | 0 | 3 | 2 | 1 |

!!

# Modular exponentiation

Often simplified using techniques such as repeated squaring.

Example: evaluate  $8^6 \bmod 11$

Method 1:

$$8^2 = 64$$

$$8^3 = 512$$

⋮

$$8^6 = 262144$$

$$\equiv 3 \bmod 11$$

overflow!

Method 2:

$$8^2 = 64 \equiv 9 \bmod 11$$

$$8^3 = 72 \equiv 6 \bmod 11$$

$$8^4 = 48 \equiv 4 \bmod 11$$

$$8^5 = 32 \equiv 10 \bmod 11$$

$$8^6 = 8^5 \cdot 8 = 80 \equiv 3 \bmod 11$$

$$\equiv 10 \bmod 11$$

lots of multiplications

Method 3:

$$6 = 110 = 1 \cdot 2^2 + 2$$

init. result  
 $r = 1$

$$110: 8^1 \cdot r^2 \equiv 8$$

$$110: 8^1 \cdot r^2 = 8 \cdot 9 \equiv 6$$

$$8^2 = 64 \equiv 9 \bmod 11$$

$$110: 8^0 \cdot (r)^2 = 36 \equiv 3 \bmod 11$$

# Modular arithmetic in PennyLane

PennyLane contains *built-in* operations:

- `qml.Adder`:  $|x\rangle \rightarrow |x + y \bmod N\rangle$
- `qml.OutAdder`:  $|x\rangle|y\rangle|b\rangle \rightarrow |x\rangle|y\rangle|b + x + y \bmod N\rangle$
- `qml.Multiplier`:  $|x\rangle \rightarrow |xy \bmod N\rangle$
- `qml.OutMultiplier`:  $|x\rangle|y\rangle|b\rangle \rightarrow |x\rangle|y\rangle|b + xy \bmod N\rangle$
- `qml.ModExp`:  $|x\rangle|y\rangle \rightarrow |x\rangle|y \cdot a^x \bmod N\rangle$

$$\frac{1}{\sqrt{2}} \left( \underset{|011\rangle}{|13\rangle} + \underset{|111\rangle}{|17\rangle} \right)$$

These implementations actually leverage the QFT under the hood; see explanation in [this PennyLane demo](https://pennylane.ai/qml/demos/tutorial_shors_algorithm_catalyst#appendix-fun-with-fourier-transforms)\*.

\*[https://pennylane.ai/qml/demos/tutorial\\_shors\\_algorithm\\_catalyst#appendix-fun-with-fourier-transforms](https://pennylane.ai/qml/demos/tutorial_shors_algorithm_catalyst#appendix-fun-with-fourier-transforms)

# Order finding

**Input:** integers  $a$  and  $N$ , where  $N \geq 2$  and  $\gcd(a, N) = 1$

**Output:** smallest integer  $r$  for which  $a^r \equiv 1 \pmod N$   
*non-zero*

Exercise: find the order of  $a = 5$  for  $N = 7$ .

$$5^6 \equiv 1 \pmod 7$$

Note: if  $N$  is prime,  $r = N - 1$   
and  $a^{-1} \equiv a^{N-2} \pmod N$

# Order finding vs. period finding

Consider the function

$$f_{N,a}(x) = a^x \bmod N$$

Order finding is equivalent to *period finding* for  $f_{N,a}(x)$ :

$$\begin{aligned} f_{N,a}(x+r) &= a^{x+r} \bmod N = a^x \underbrace{a^r}_{1} \bmod N \\ &= a^x \bmod N \\ &= f_{N,a}(x) \end{aligned}$$

# Order finding on a quantum computer

$$f_{N,a}(x) = a^x \bmod N$$

Define a unitary operation that performs

$$U_{N,a} |k\rangle = |(a k) \bmod N\rangle$$

If  $r$  is the order of  $a$  modulo  $N$ , and we apply  $U_{N,a}$   $r$  times,

$$(U_{N,a})^r |k\rangle = |(a^r k) \bmod N\rangle = |k\rangle$$

# Order finding on a quantum computer

*We'll start here next time.*

$U_{N,a}$  has the same order,  $r$ . We can use QPE to efficiently find  $r$  with a clever initialization of the target register.

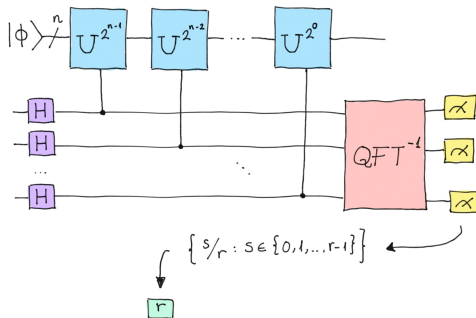


Image credit: Xanadu Quantum Codebook node S.3

# Next time

## Content:

- Factoring with Shor's algorithm

## Action items:

- ① Tutorial Assignment 3 due Friday 23:59
- ② Midterm checkpoint report due Friday **5:00pm**

## Recommended reading:

- Codebook modules QFT, QPE, SH
- Nielsen & Chuang 5.3, Appendix A.5
- PennyLane demo: Quantum just-in-time compiling Shor's algorithm with Catalyst